

Privacy is Not One-Click: Designing Robots That Adapt to Older Adults' Changing Boundaries

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Abstract

Social robots in the home bring new privacy risks and concerns for older adults. Yet, current technology privacy mechanisms typically use a one-time and universal consent mechanism (e.g., user agreement checkbox, browser cookie setting, etc.), lacking consideration of how privacy is holistically experienced. Designing for privacy requires a multidimensional approach to support how older adults experience privacy. To investigate older adult-centered privacy mechanisms for social robots, we conducted two participatory design (PD) workshops at local assisted living facilities. Our findings from these workshops suggest that older adults do not treat privacy as static, but as a temporal and situational practice that requires continuous negotiations and revisions. We subsequently conducted a post-PD speculative design (SD) process that extracted three design features for privacy—aware social robots-privacy profiles, real-time privacy feedback, and data ownership tools—that can support older adults' multidimensional privacy experiences.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**; **Empirical studies in HCI**.

Keywords

privacy, social robots, older adults, speculative design, participatory design

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1 Introduction

As the global population ages, social robots are increasingly explored as a technology-based solution to support older adults' daily activities [71, 77], health monitoring [16, 50], and social well-being [62, 72]. To achieve this, robots are often equipped with various sensors (e.g., cameras and microphones), collect personal and environmental data, and transfer data to and from cloud-based storage and services. For older adults, who often have limited technology backgrounds, such continuous monitoring and data collection in intimate private spaces and settings can raise acute privacy and trust concerns [54, 55].

Robots engage humans in physical, informational, psychological, and social dimensions [7, 57], contributing to privacy's multifaceted nature. These dimensions interact in complex ways that shape older adults' overall privacy experience [65] and require dynamic changes in regulating boundaries as privacy concerns and needs fluctuate based on specific contexts [2]. Because of pronounced privacy boundaries introduced by robots, technology literacy is required to understand and manage a robot's data security and privacy. Lack of literacy can lead users, including older adults, to accept default settings or avoid adjustments altogether [86]. When older adults interact with privacy settings, this can limit their abilities to manage privacy risks [24] and create feelings of constantly being surveilled and deprived of privacy [66].



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Despite growing research at the intersection of aging, robotics, and privacy, significant gaps remain, as prior studies often treat physical, informational, psychological, and social privacy as isolated concepts rather than a multidimensional whole. For instance, context-aware systems that disable cameras in sensitive areas such as bedrooms or bathrooms primarily aim to support physical privacy [34, 40], but may neglect psychological concerns (e.g., whether users feel watched), or social dynamics (e.g., how others in the space are affected). Similarly, transparency solutions, such as verbal notifications or sensor indicators, aim to improve awareness and address informational or social privacy [2, 69], yet they rarely provide users with real agency over the robot's behavior or address concerns around spatial or psychological intrusion. As a result, these mechanisms remain fragmented, addressing only certain privacy types and elements in isolation. Since users, including older adults, experience privacy concerns holistically during interactions, it is critical to consider the multidimensional nature of privacy when designing privacy features.

Despite these complexities and nuances surrounding privacy, older adults are frequently excluded from technology design processes, including defining technical and feature requirements for social robots [55]. Privacy, especially in domestic and care contexts, is shaped by personal routines, evolving relationships, and social situations. Thus, having autonomy over one's privacy requires a relational and situational process and cannot be achieved by a one-size-fits-all technical function or feature. Through participatory design (PD) workshops and the incorporation of speculative design (SD) into the design process, our work engages with the following research questions:

- **RQ1:** How do older adults conceptualize privacy across physical, informational, psychological, and social dimensions when interacting with social robots?
- **RQ2:** What design solutions do older adults envision for preserving privacy across these dimensions?
- **RQ3:** How can SD be incorporated into the design process post-PD workshops to support interpreting older adults' privacy expectations and privacy designs for social robot interactions?

In this paper, we highlight two phases of our work: Phase 1 describes the PD workshop and insights learned from collaborating with older adults; and Phase 2 details how we engaged SD to extend the privacy solutions developed in the PD workshops. Overall, we contribute the following: (1) empirical insights into how older adults conceptualize and negotiate privacy across physical, informational, psychological, and social dimensions in the context of social robots; (2) a speculative post-PD process to support analyzing PD data to surface situated privacy expectations; and (3) a set of privacy design features for implementing multi-dimensional privacy mechanisms grounded in older adults' lived experiences.

2 Background

2.1 Multidimensionality of Privacy in Older Adult-Robot Interaction

Privacy is a multifaceted concept often defined along several dimensions – physical, informational (data), psychological, and social

[7, 38, 57]. In the context of human–robot interaction (HRI), these different dimensions of privacy intersect in unique ways because robots are typically equipped with various sensors that may collect physical, informational, and psychological data to perform diverse tasks in personal environments [7, 57].

Physical privacy involves the protection of bodily and spatial boundaries, shielding individuals from unwanted observation or intrusion [38, 45]. With their physical embodiment, social robots often need to have close proximity to the user to help with daily tasks or monitoring, which can intrude upon personal space [41, 67, 70]. When interacting with technologies that can monitor users, older adults also adjust their behaviors to preserve and guard their privacy and dignity through privacy-enhancing behaviors, such as covering the camera's lens with an object, hiding personal items, and even censoring their speech to avoid revealing sensitive information [9].

Psychological privacy relates to the preservation of one's mental and emotional space, enabling autonomous thought and expression without undue surveillance [38]. The need for mental solitude and freedom from perceived observation is paramount in older adults' interactions with social robots, reflecting a desire to relax without maintaining a social façade or adhering to social norms, such as dressing formally [14, 41]. Older adults have demonstrated that when interacting with social robots, they engage in activities that protect their psychological privacy, such as only turning on the robot for structured activities or companionship, but turning it off for private moments [10, 54].

Social privacy concerns the regulation of interpersonal interactions, including decisions about who may engage, communicate, or observe one's behavior [7, 57]. Social robots often interface with multiple stakeholders (e.g., older adults, family members, caregivers, and peers), raising important questions about how the robot manages social interaction boundaries on behalf of the older adult [2, 34, 47]. Another aspect of social privacy is how the robot handles older adults' personal information in social contexts [19]. A social robot may possess sensitive information about its owner, such as health status or daily habits, that could surface during interactions with others, making the decision of what it should disclose a delicate social privacy issue [19]. These elements of privacy are often negotiated with multiple stakeholders in older adults' social and care networks [49].

Informational privacy refers to control over the collection, use, and dissemination of personal data [25, 38]. This type of privacy is a major concern for older adults considering social robots in the home [39, 46, 83]. Robots equipped with cameras, microphones, and other sensors can capture intimate details of daily life, raising concerns among older adults about who has access to this sensitive data. [46]. Older adults value transparency and agency as core values for social robots that will handle sensitive information [2], with privacy being ranked as a top concern [15]. Older adults have advocated for robot privacy features, such as the ability to refuse or pause monitoring and transparency around when the robot is recording information, to support their knowledge of the robot's sensing capabilities [15, 37]. Beyond data collection, older adults are often concerned about the misuse of personal information, including fears around hacking, identity theft, or information being leaked to third parties [37].

2.2 Privacy Design Strategies for Social Robots

Researchers have proposed a variety of privacy-preserving design strategies in HRI and human-computer interaction (HCI) [2, 8, 17, 40, 42, 65]. These solutions fall broadly into three categories: (1) context-aware adaptations that dynamically tailor robot behavior based on situational cues (e.g. [2, 8, 32, 34, 40, 74]), (2) data protection mechanisms that safeguard users' personal information post-collection (e.g. [29, 42, 53, 63, 64, 74]), and (3) transparency and notification systems that inform users and bystanders about the robot's sensing and data practices (e.g. [2, 64, 69, 70, 84]).

2.2.1 Contextual Privacy Solutions. Context-based solutions recognize that privacy expectations are highly context-dependent [2, 8, 32, 34, 40, 74]. Privacy solutions that adapt to context aim to make the robot aware of situational factors and social norms, enabling it to modify its behavior to better respect users' privacy. This might include automatically disabling, blurring, or obfuscating the robot's camera images upon entering sensitive areas (e.g., bedroom) [8, 34, 40] or by adjusting a robot's data collection/recording status based on by the sensitivity of information measured via users' sentiment, social relationships, and conversation topics [74]. These contextual privacy solutions use context-specific rules to match users' nuanced privacy expectations in various settings, with the assumption that people's expected privacy may shift across time and social roles [56].

Despite their strengths, context-based approaches have notable limitations. For example, a robot might refrain from recording in one context but still collect extensive personal data in another permissible context, raising questions about how that data is stored or shared. Similarly, even if a robot is inactive in a private moment, a user might still feel watched or worry that the robot could misinterpret context. Therefore, context-aware solutions may preserve physical and social privacy in the moment, but they leave gaps around downstream informational control and users' psychological comfort.

2.2.2 Data Protection Mechanisms. While context-based privacy solutions try to prevent privacy intrusions at the interaction level, data protection mechanisms focus on security and management of the collected data. These solutions aim to protect informational privacy by ensuring users' control of their personal data after the collection is completed [29, 42, 53, 63, 64, 74]. Fundamental principles in this area are data minimization (robots should only collect minimal data required for their functionality) [42, 64] and ensuring that users can decide what a robot records and who is permitted to access that information [2, 53]. Many privacy frameworks for robots insist that a robot "should not record, store, or transmit personal data without explicit user consent" [2, 53] and control mechanisms should support only authorized individuals to retrieve or view stored data [53], e.g., profile- and permission-based data sharing and storage [74] to prevent inadvertent disclosure of one user's information to another. However, technical data protections alone do not guarantee psychological privacy. Even when data is securely stored, users may still feel surveilled if a robot's presence is perceived as invasive if the robot's status is not clearly communicated [19].

2.2.3 Privacy Notifications. Transparency in the context of privacy means ensuring the robot's capabilities, data practices, and decision-making are visible and understandable to users and bystanders [70]. Because robots are embodied and autonomous, people can easily misunderstand what the robot is doing, leading to uncertainty or misconceptions about privacy [31]. Consequently, a major line of privacy solutions centers on notifying users about the robot's sensors and data practices, enabling them to manage their privacy effectively [2, 31, 64, 69, 70, 84]. This can include design features such as lights or sounds that are activated when devices are listening [31] (supporting physical and informational privacy), notices posted in public spaces where robots are operating [64], and verbal cues to signal when data is being stored or collected [64]. Through these features and others, social robot design can communicate robots' capabilities to users—such as whether it can recognize faces, store conversations, or share data with cloud services [64], treating privacy as an ongoing dialog with the user, rather than a hidden background process [65]. A significant gap lies in traditional privacy notices, often relying on complex legal language that most older adults find difficult to understand [17], underscoring the need for more intuitive privacy interfaces in social robots. Overall, our work seeks to address the need for holistic multi-dimensional privacy mechanisms in social robot design that are grounded in older adults' privacy conceptualizations, concerns, and design ideas.

2.3 Speculative Design Related to Participatory Design

SD, or critical design, has been incorporated into HCI and HRI spaces to provoke reflection and envisioning of alternative futures around technology [21, 26]. Previous work used SD not only to engage participants to co-design potential futures [36, 76] but also to aid researchers to learn about how participants perceive, engage with, and reflect on SD experiences and/or design artifacts [3, 6, 13, 22, 30, 48, 51, 81, 82]. Privacy-based research has also used speculative prototypes, scenarios [80, 82], speculative probes, fictions, and workbooks to investigate and encourage discussion around privacy [11, 28, 43, 58–61, 81, 81, 82]. Wong & Mulligan [80] note that more HCI design researchers should collaborate with privacy-focused HCI researchers, engaging a broader set of stakeholders through SD methods to identify privacy-centered solutions to more effectively define privacy problems grounded in social values.

In addition to expanding how SD is incorporated into privacy-based designs, HCI and HRI researchers have also reconsidered how and where in the design process SD could be incorporated. Instead of engaging participants in SD experiences or interactions, researchers have used SD as a method for engagement and reflection by the research team [12, 18, 35, 52, 75, 78] to "identify future directions for design" [12, 18, 75]. Our work expands this conceptualization and use of SD by building upon data gathered from PD workshops with older adults, which, to our knowledge, has not been done before. Speculative design has been incorporated directly within PD processes previously [30, 44]. However, our design exploration approach focuses on supporting PD processes after workshops through SD, knitting together and interconnecting older adults' concerns and design ideas around privacy with the research

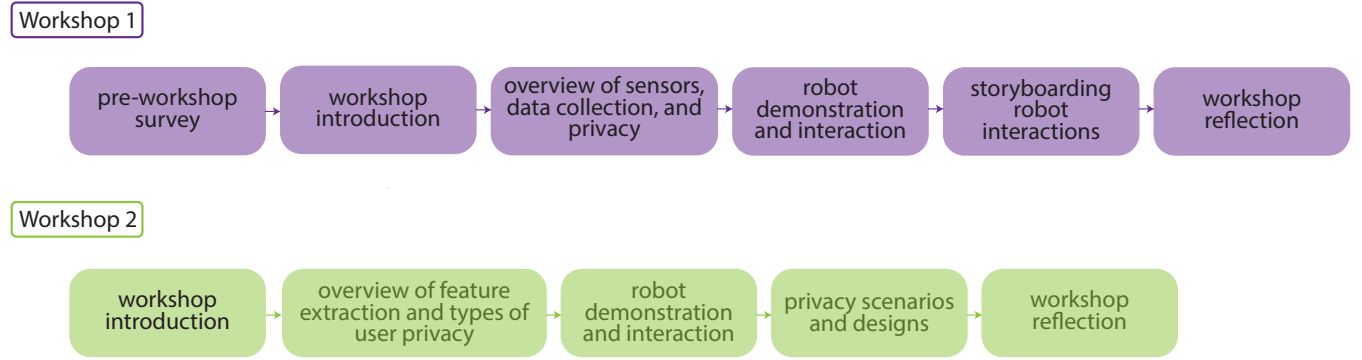


Figure 1: Overview of the two-part participatory design workshop structure. Workshop 1 focused on introducing robot sensors and their capabilities and storyboard creation; Workshop 2 involved privacy scenario reflection and discussion on privacy-preserving designs with older adults.

team’s knowledge and experience of social robot design to realize design features that can be implemented in the next design phase.

3 Researcher Positionality

In this work, we engage in a PD process with older adults, followed by a SD process with the research team, building upon and expanding older adults’ designs. The research team’s analysis of the PD workshops and SD development process was influenced by the team’s expertise, research areas, and experiences. The research team is composed of a variety of disciplinary backgrounds, including engineering, computer science, artificial intelligence, HRI, human-centered design, science & technology studies, and disability studies. This combined expertise supported the SD process and generation of design features, informed by the PD analysis.

4 Phase 1: Participatory Design Workshops for Social Robot Privacy Solutions

We conducted two PD workshops, each lasting approximately three hours, to gain a deep understanding of older adults’ lived experiences regarding technology-related privacy and identify practical design solutions for future social robot privacy mechanisms. Rooted in empowering and valuing the lived experiences of stakeholders, PD can be a powerful approach to inform privacy interfaces that are designed to be older adult-centered. PD centers older adults’ experiences, concerns, and ideas through active co-creation [55]. This approach is particularly critical in the context of privacy, which is not a universal or static concept but one that is deeply personal, contextual, and shaped by lived experience [56]. Participatory methods are thus uniquely positioned to uncover these situated concerns by directly involving older adults to articulate their needs and values. Previous work has engaged older adults in PD of robots, unearthing privacy concerns and insights into how older adults experience privacy in relation to technology [54, 55]. Older adults emphasize how privacy is often entangled with physical routines, social dynamics, and spatial constraints, underscoring its central role in technology design and illustrating how PD can explore these concepts [73, 85].

To uncover older adults’ expectations and design for privacy in social robot interactions, we conducted two PD workshops with

older adults in local assisted living facilities. During these workshops, older adults engaged in design activities that generated and probed privacy configurations. All study procedures were reviewed and approved by Purdue University’s Institutional Review Board. Informed consent was gathered from all participants.

Table 1: Summary of participants and the workshops they attended.

	Workshop 1 Participation	Workshop 2 Participation
Study Site #1	p01, p02, p03, p04, p05, p06, p07, p08, p09	p01, p02, p03, p04, p05, p06, p07, p08, p09
Study Site #2	p10, p11, p12, p13, p14	p10, p11

4.1 Participants

As our work’s scope is not to compare the two sites, we present the combined participant demographics of all fourteen participants. We recruited fourteen older adult participants (7 female and 7 male) from two local assisted living facilities. The inclusion criteria for participants were (1) 65 years of age or older, (2) native speakers of English or fluent in oral/written English, and (3) self-reported normal or corrected-to-normal hearing and vision. Eight older adults participated in the first study site: 8 in Workshop 1 (W1); 8 in Workshop 2 (W2) (same participants at both workshops). Five older adults participated in the second study site: 5 in W1; 2 in W2 (the 2 participants in W2 also completed W1). Participant attendance is summarized in Table 1. All participants, regardless of site, completed the same workshop activities.

4.2 Methodology

The workshops were held at each local study site to support older adults’ easy access. An overview of the workshop structure and respective activities is illustrated in Figure 1.

4.2.1 Workshop 1: Understanding Robot Sensors and Imagining Robots for Older Adults. In W1, participants explored how social



Figure 2: (a) Four commercial social robots used in the workshops (clockwise from top left): Astro, Misty, Emo, and Loona; (b) Participants interacted with commercial social robots during the first workshop; (c) Examples of daily life scenarios from six commonly proposed application domains for social robots.

robots sense and collect different types of data through interactive demonstrations of commercially available robot platforms and storyboarding. At the beginning of the workshop, participants completed a short pre-workshop survey to capture their demographic background, prior experience with technology, and initial attitudes toward robot-mediated data collection and privacy. To prepare participants for the hands-on design activities that followed, the research team gave a presentation that introduced various sensors used in robots, such as cameras, microphones, and GPS, and explained how these support robot perception of the environment, human behavior, speech, presence, and health signals to interact with people in real-world settings.

Following the presentation, older adult participants interacted with four commercial robots (Misty¹, Amazon Astro², Emo³, and Loona⁴) to experience how the sensor modalities discussed in the presentation are used in the robots (Figure 2(a)). Astro, Misty, Emo, and Loona were selected as they represent social robots with varying functionalities and embodiments. Astro can map and navigate one's home autonomously and socially interact with people through verbal interactions. Misty is a research robot platform that has a built-in Wizard-of-Oz interface that allowed our study participants to directly control the robot and visually explore how different robot sensors collect and process multi-modal data. Emo and Loona are marketed as children's toys and showcase different ways that a social robot could interact both orally and physically with people for entertainment. This hands-on interaction helped older adults understand how different sensors (e.g., camera, microphone, touch sensor) enable the robots to actualize various core functionalities (e.g., face recognition, object detection, audio/video streaming, voice recognition, responding to touch). Participants interacted with the robots for twenty minutes by using voice commands, requesting songs or jokes, and/or directly controlling the Misty robot through the system's Wizard-of-Oz teleoperation interface (Figure 2(b)). The workshop facilitators actively invited the participants to ask questions about what data each sensor collected, how the data might be stored, and why the data mattered in robot functionalities.

Building on these experiences with the social robots, the research team presented a brief case study of a data leak incident involving the Roomba robot vacuum [27]. This case study was used to contextualize potential risks and ethical stakes of data collection through robot sensors and to encourage participants to reflect on the everyday implications of such systems. Although Roomba is not designed as a social robot, we chose to use this case study because many participants were familiar with the Roomba robots as they are more common home robot, and the real-world case of a familiar robot's data leakage would resonate well with our older adult participants and help them speculate how social robots could potentially impact their privacy.

After these foundational activities, we engaged older adults in a storyboarding activity to imagine potential robot roles and usage. The storyboarding activity supported older adults in detailing their expectations and concerns about robotic sensing in the home, grounded in realistic expectations of privacy risks. Storyboards are a widely used design tool in HCI and HRI to help non-experts visually articulate speculative or lived experiences with technology. To support storyboard ideation, we presented 12 images showing everyday activities drawing from 6 common applications of social robots (medical adherence, emotional wellness, memory support, exercise coaching, physiological monitoring, and social connection) based on prior PD research [55] (Figure 2(c)). Participants selected one of the images and created a storyboard of how they imagined a robot could support them in this activity (example storyboard worksheet shown in Figure 3(a)). Participants were asked to identify the sensors and data required to realize the interaction in the storyboard and what the expected benefits were, often explaining technical affordances with everyday goals and privacy expectations. Facilitators circulated during the activity to answer questions, support storyboard creation, and gently prompt deeper thinking (e.g., "What might the robot need to sense here?" or "How would you want to be notified if something was stored?"). Overall, W1 equipped participants with foundational knowledge of robotic sensing and empowered them to creatively explore how such technologies could be meaningfully and responsibly integrated into their lives.

¹<https://www.mistyrobotics.com/>

²<https://www.amazon.com/Introducing-Amazon-Astro/dp/B078NSDFSB>

³<https://living.ai/emo/>

⁴<https://keyirobot.com/products/loona>

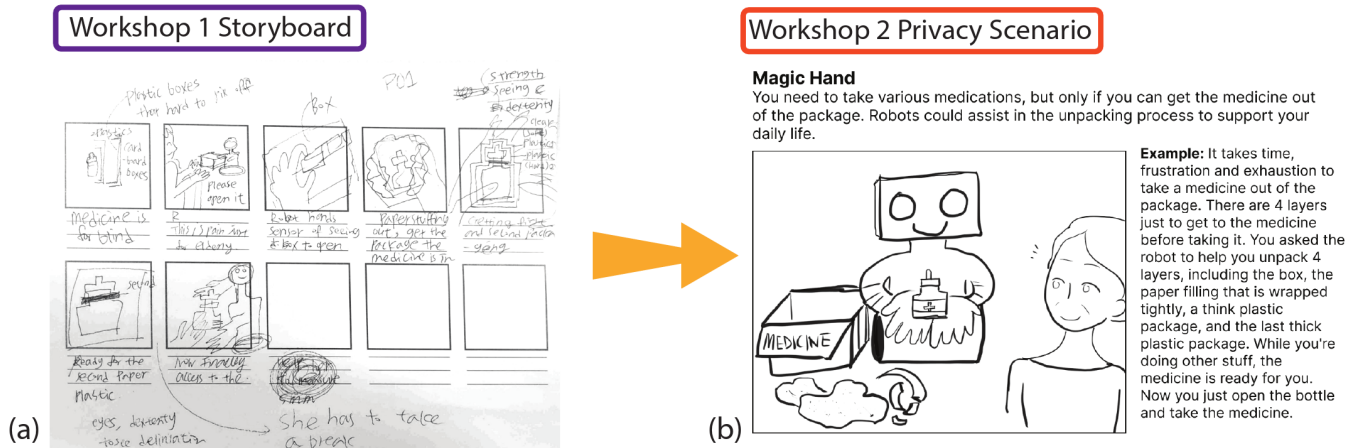


Figure 3: Based on the W1 storyboards, robot-related privacy interaction scenarios were generated to be discussed in W2.

4.2.2 Workshop 2: Co-designing Privacy Solution for Robots and Thematic Synthesis. W2 focused on identifying potential privacy risks and co-designing potential privacy solutions for robots. The workshop session built on storyboards developed in W1. Prior to W2, the research team analyzed the W1 storyboards and distilled them into representative privacy scenarios (similar to Wong et al. [82]). This resulted in three privacy scenarios: (1) location tracking for fall alerts, (2) speech monitoring for cognitive assistance, and (3) continuous physiological logging. Each scenario provided a brief description and example of how the scenario may occur in older adults' lives (Figure 3(b)). The description supported making abstract sensing processes more concrete and to scaffold participants' thinking about privacy risks.

W2 began with the researchers presenting a brief overview of the sensor-to-service dataflow pipeline. This highlighted how cameras, microphones, and physiological monitors generate raw streams of data that are processed, stored, and potentially shared to enable certain robot features. This provided a groundwork and foundation for a hands-on privacy design exercise. Groups of 3-4 people were assigned a privacy scenario to critically examine and identify potential privacy risks. Facilitators briefly introduced common categories of privacy risks (e.g., unbounded data retention, inference of sensitive conditions, unauthorized third-party access) to scaffold participants' thinking without prescribing specific concerns. After identifying potential privacy risks in each scenario, participants engaged in small-group discussions to generate potential strategies and solutions to mitigate these risks. Facilitators prompted each group to consider how the privacy scenarios could be redesigned to better align with user expectations for control, transparency, and trust.

The goal of this activity was to generate tangible and user-centered design ideas that addressed the identified risks in practical and contextually relevant ways. Groups were encouraged to explore both technical and interaction-level interventions, such as physical indicators, data storage preferences, and consent mechanisms, and to think creatively and critically about how these features might function in everyday use. Participants documented their proposals using sketches, annotations, and storyboard revisions. These were

then shared in a full-group debrief to reflect on recurring themes and design implications. Overall, W2 supported older adults in identifying privacy risks and ideating potential privacy design interactions.

4.3 Data Collection and Analysis

The main source of data for this paper included: (1) audio-recorded conversations from both workshops, (2) participant-created storyboards from Workshop 1, and (3) workshop observation notes documenting participant engagement, questions, and emergent themes. The transcribed audio recordings from both workshops were checked for accuracy and analyzed through a thematic analysis approach [5]. First, four researchers individually extracted preliminary and sub-themes from the transcripts (two researchers for the first study site transcripts; two researchers for the second study site transcripts). The research team then discussed, compared, and consolidated these themes in a second pass that was conducted by the same four researchers. Discrepancies were resolved through team discussions.

To analyze the storyboards, the research team conducted a qualitative content analysis [20] focusing on: (a) the types of sensors depicted; (b) the kinds of data imagined to be used or collected through these sensors (e.g., video, audio, physiological signals); (c) any contextual cues indicating when and why sensing occurred; and (d) any user-facing benefits, concerns, or implied privacy expectations. Coding was guided by both deductive privacy concepts (e.g., consent, data minimization, etc., drawn from [42, 56]) and inductive themes that emerged from participants’ drawings and annotations. Storyboard elements were then cross-referenced with themes from the transcripts to triangulate participants’ values, concerns, and design intentions. This analysis informed the W2 privacy scenarios (as previously described in Section 4.2.2).

4.4 Results

This section describes results on older adults' baseline technology-related privacy concerns and preferences (pre-workshop survey), older adults' storyboarding (W1), the robot sensors included in

Table 2: Pre-workshop assessment on participants' perceived comfort on various data collection ($N=14$, "How comfortable are you with the following types of data being collected?")

Data Collection Type	Very Uncomfortable	Somewhat Uncomfortable	Neither Comfortable nor Uncomfortable	Somewhat Comfortable	Very Comfortable
Visual Data Collection (i.e., face lock)	2 (14.29%)	3 (21.43%)	3 (21.43%)	1 (7.14%)	4 (28.57%)
Audio Data Collection (i.e., voice recognition)	2 (14.29%)	4 (28.57%)	2 (14.29%)	2 (14.29%)	4 (28.57%)
Internet Browsing Data (i.e., internet search history)	4 (28.57%)	4 (28.57%)	3 (21.43%)	0 (0%)	3 (21.43%)
GPS Location Data (i.e., geographic location of your device)	3 (21.43%)	4 (28.57%)	1 (7.14%)	3 (21.43%)	3 (21.43%)

Table 3: Pre-workshop assessment on participants' privacy-related concerns and preference for data collection ($N=14$).

	Strongly Agree	Somewhat Agree	Neither Agree nor Disagree	Somewhat Disagree	Strongly Disagree
I have been concerned about my privacy when using technology.	5 (35.71%)	5 (35.71%)	2 (14.29%)	1 (7.14%)	0 (0%)
I want to have control over my data being collected by technologies.	7 (50.00%)	5 (35.71%)	1 (7.14%)	0 (0%)	0 (0%)
My devices inform me about how they are collecting data and they are storing it.	2 (14.29%)	3 (21.43%)	4 (28.57%)	2 (14.29%)	2 (14.29%)
I understand how my data is being collected, stored, and used in my technologies.	3 (21.43%)	2 (14.29%)	0 (0%)	3 (21.43%)	5 (35.71%)
I do not mind technology collecting my data because the technology works for me.	2 (14.29%)	6 (42.86%)	1 (7.14%)	2 (14.29%)	2 (14.29%)

older adults' storyboards (W1), and older adults' privacy concerns and proposed solutions (W1 + W2). The results on future robot scenarios (Sec. 4.4.2) and common sensors (Sec. 4.4.3) describe the results of the content analysis of the storyboards. The results on older adults' privacy concerns and proposed solutions were revealed through the thematic analysis of the workshops with the themes detailed in Sec. 4.4.4, Sec. 4.4.5, and Sec. 4.4.6 (themes are shown in bold text).

4.4.1 Pre-workshop: Participants' Baseline Privacy Concerns and Preferences. Older adult participants used a variety of technologies

in their daily lives, including smartphones (92.86% of participants), smart speakers (35.71%), laptops (78.57%), social media (78.57%), iPads/tablets (50.00%), and other technologies (35.71%, such as smart televisions and smart watches). These technologies were used for tasks, such as communication and socialization (92.86%), access to information and news (100.00%), entertainment (85.71%), online shopping (71.43%), communicating with doctors and other people in an older adult's healthcare team (78.57%), supporting health or wellness, e.g., meditating, exercise (28.57%), and other applications (64.29%), including making things easier to read, gaming, creative tasks such as drawing, finding recipes, and map-GPS. Older adults'

mixed comfort levels with various data collection types, concerns around privacy, and mixed opinions around understanding their devices' data collection are further detailed in Tables 2 and 3.

4.4.2 Future Robot Scenarios for Older Adults. During the storyboarding exercise, older adults ideated potential scenarios and use cases that a social robot could help them complete or achieve daily tasks (examples of created storyboards shown in Figure 4). Of the ten storyboards, six included drawings, while four were solely written descriptions. Both visual and written storyboard formats reflected participants' preferences well. Four participants chose not to produce a storyboard because they deemed no utility of robots in their lives. Participants' intended uses for social robots leaned more on practical usage than social support, and fell into three main categories: (1) general assistance, (2) cognitive/mental support, and (3) physical support.

Participants' storyboards often described *general assistance*, where robots provided support with limited social interaction (storyboards were named by the research team). Multiple scenarios included robots being used as a "Walking Buddy" on outdoor walks to provide navigation to set locations (P06), safety at night (P03), and information on nature (P07). In these cases, the robot could "point out the path and trees in the way" and alert when users might "bump into something" (P03). Other participants wished for a personal assistant who could complete daily chores and bring items such as canes, TV remotes, and medication when asked (P05, P08). Finally, a participant felt a robot could provide general support when they engage in puzzle activities, such as crosswords, by providing clues and visual support, e.g., magnifying the text (P02).

Participants also wanted social robots to provide *physical support*, aiding with tasks that require dexterity. P02 suggested that a robot could write the answers for crossword puzzles based on their verbal input to prevent their hands from getting "cramped." Additionally, P01 suggested a robot could act as a "Magic Hand" and assist them in opening multi-layered packages, which was described as "one of the biggest headaches" due to P01's "arthritis, tendinitis, and blindness." They suggested that an ideal robot should offer assistance to older adults' diverse accessibility needs.

Participants indicated that social robots could provide *cognitive support*, such as memory assistance. P05 suggested that a robot could be used as a "Life Record," a catch-all memory tool to remember every aspect of one's daily life, including people's faces, details of conversations one had with others, and health records. P05 imagined the Life Record robot would assist them in remembering the location of misplaced objects (e.g., "Where did I put my keys?") or recall information about specific people (e.g., "Where did I meet that person?"). P04 also wished for a robot that could provide reminders for medical needs (e.g., checking blood pressure or taking medication, "changing [their] insulin cartridge") to increase their medical adherence to the prescribed regimen.

4.4.3 Common Sensors in Older Adult-Robot Interaction Scenarios. Cameras emerged as the most ubiquitous sensor across all scenarios, with participants recognizing their necessity for multiple functions. In P03's scenario, cameras enabled the robot to recognize when the user was walking unsteadily and identify obstacles in the path ("he [the robot] walks with me. So you'd have to know the location and the

camera...it would point out the path and the trees in the way. So use location, voice, and sight."). For P05's personal assistant role, cameras were essential for recognizing family members versus strangers, enabling the robot to provide different levels of access to personal information. Participants also identified cameras and touch sensors as crucial for dexterity support scenarios, where robots needed visual feedback to locate and manipulate objects such as medication bottles or dropped items. In addition, *navigation-related sensors* (e.g., GPS, depth sensors, and motion sensors) were consistently identified in scenarios requiring physical interaction and movement in the environment. P03 identified depth and location sensors to be necessary for the robot in their storyboards to provide navigation support and identify potential obstacles in the path. P04 incorporated motion sensors to detect falls while preserving privacy by avoiding constant visual surveillance. Participants often identified *microphones* as the required sensor when envisioning voice-based interactions with the robot. P02's crossword support specifically required microphones for the robot to hear letter clues and respond with suggestions when the user's hands were occupied with writing. Emergency response scenarios across multiple storyboards identified technologies required for communication (e.g., microphones, WiFi, and Bluetooth connectivity) to detect distress calls or changes in older adults' health status.

This sensor mapping process revealed that the educational content in W1 offered foundational knowledge for older adult participants to gain intuitive understanding of various robot sensors and data collection mechanisms to further discuss privacy concerns and solution proposals during the rest of the workshops.

4.4.4 Preference for Context-aware Privacy Controls. Participants noted their desire for robots that could fluidly adapt their privacy settings based on moment-to-moment changing contexts in everyday life. They viewed that most current technologies/devices treat privacy as a static feature locked as a fixed and one-size-fits-all privacy setting ("Every time I turn on a device, it asks me if I want to share something, and the default is always 'yes.'" (P03)), and imagined that *context-aware* robots would adjust their sensing status and data sharing behaviors in real time, whether during moments of solitude, emergencies, or social interactions. Participants also described several privacy concepts that could support these context-aware privacy controls, including a location slider and mode-based privacy presets. A location slider is an intuitive interface for scaling location sharing granularity (e.g., street-level vs. neighborhood radius), offering older adults finer control depending on their activity. Mode-based privacy presets were identified as predefined privacy configurations that are linked to daily activities, such as "Walking," "Resting," or "Emergency," to allow users to toggle modes with a simple voice command or touchscreen interaction.

4.4.5 Desire for Tangible and Physically Verifiable Privacy Controls. While participants described interfaces based on privacy controls, they also expressed widespread skepticism towards the opacity of digital interfaces and cloud-based systems ("Big daddy in the sky" (P09)) and showed a preference for tangible and physically verifiable privacy controls that allow direct oversight of the data lifecycle. As P09 remarked, "If it [the robot] can download firmware, it can upload whatever's [data] there," while P08 emphasized local control, stating "My old camera stored photos on the memory card. I could take it out



Figure 4: Storyboards created during the participatory design workshops (named by research team): (a) "Walking Buddy" (P03): A robot accompanies an older adult on a late evening walk and provides navigation support via GPS, identifies potential obstacles on the path via camera, offers light for safe walking via its flashlight, and offers fall detection in case of an emergency; (b) "Magic Hand" (P01): A robot provides assistance to older adults with fine motor impairment by opening a multi-layered medicine package; (c) "Blood Pressure Cuff" (P09): A robot helps its user take their blood pressure when they are forgetful by bringing the necessary equipment and learning the appropriate procedures involved.

and lock it in my drawer." These concerns underscore the need for privacy mechanisms that are not only visible and trustworthy but also grant users meaningful control over their personal information.

In addition, **on-device data processing and storage** emerged as a critical priority, with participants advocating for local data handling to avoid cloud dependency. This sentiment was reinforced by proposals for modular hardware solutions, such as easily swappable memory modules that users could physically manage. As P09 articulated, "Can you slide new memory into them? How long is that stored? And where is it stored? What does she lose when you turn her off?", highlighting demands for transparent storage mechanisms.

Furthermore, on-device processing was strongly favored to avoid any external data transfers entirely.

Furthermore, some participants showed a preference for **modular storage media** that enable physical destruction for guaranteed deletion, over embedded storage devices that cannot be separated from the robot. P05 proposed that one should be able to "physically destroy it [memory card] through an industrial shredder", contrasted sharply with digital erasure methods viewed as theoretically reversible. The destruction imperative extended to adaptive privacy personalization, with participants rejecting standardized data deletion timelines in favor of personalized destruction schedules: "If

Table 4: Connection of each design feature designed by older adults in the PD workshops to physical, psychological, social, and information privacy dimensions.

Design Features	Physical Privacy	Psychological Privacy	Social Privacy	Informational Privacy
<i>Context-aware privacy setting adjustment</i>	Adjusts sensing, location access, or proximity behaviors based on room type or activity to avoid intrusion or increase sensing.	Reduces perceived surveillance by adjusting sensing intensity or disabling certain functions based on contextual appropriateness.	Modifies data capture when guests or non-users are present, preventing accidental recording and supporting shared norms of interpersonal privacy.	Dynamically scales data collection, storage, granularity, and transmission depending on situational risk or potential support need.
<i>On-device data processing and storage</i>	N/A	Reduces anxiety about constant monitoring and loss of control, giving users a sense of containment.	Prevents unintended sharing of sensitive behavior with caregivers, family, or third parties.	Strongly supports data minimization by storing data locally and reducing risks of leakage, profiling, or unauthorized access.
<i>Modular storage media</i>	Allows users to physically remove data when needed, supporting boundaries around personal environment.	Enhances feelings of agency and control by enabling “tangible” deletion or safeguarding of sensitive data	Users can curate which data are available to whom by physically transferring or withholding storage modules.	Enables selective retention, deletion, or transfer of data, strengthening user control over long-term data traces.
<i>Hardware-based privacy indicators</i>	Provides clear cues when sensors are active, allowing users to protect private spaces or bodily activities.	Reduces psychological burden and uncertainty by making surveillance states visible and understandable.	Signals to bystanders when recording is happening, preventing accidental capture in shared spaces.	Offers transparent data-capture states, supporting consent and reducing inadvertent data collection.
<i>Opt-out-by-default architecture</i>	N/A	Supports mental comfort by preventing assumptions of constant monitoring and respecting user autonomy.	Prevents inadvertent participation of visitors or household members who do not wish to engage with the robot.	Minimizes default data collection and only data explicitly permitted by users are gathered, reducing unnecessary exposures.
<i>Dialogue-based permission systems</i>	N/A	Provides supportive, easy-to-understand conversational explanations that build trust and reduce cognitive load.	Encourages socially appropriate consent practices when others are present (“May I record this?”)	Enables granular, comprehensible consent for specific data types, uses, or storage duration, strengthening informed data control.

I had control over that, if I could go in every night and erase the data, or once a week go in and erase all that data and I know it's not stored anywhere, then I would have more confidence in the tool or the device” (P06). This preference reflects a desire to synchronize data management with personal routines, matching older adults' comfort levels.

Hardware-based privacy indicators were proposed as essential safeguards, with participants preferring mechanical interfaces that provide unambiguous *realtime privacy notifications*. P06 emphasized the need for visible recording indicators, stating, “*There should be a light that turns red when it's recording-like old cassette tapes*”, drawing parallels to familiar analog technologies. Participants suggested integrating “*physical switches for microphones*” (P05) and “*mechanical lens covers for cameras*” (P08) that provide both functional control and psychological reassurance through their tangibility. These proposals reflect a broader distrust of software-based privacy notice indicators, which participants described as easily manipulated or concealed by manufacturers.

4.4.6 Need for User-centric Consent Models. Participants re-imagined consent as an ongoing negotiation rather than a one-time agreement. This paradigm shift emerged from widespread distrust of

conventional “set-and-forget” privacy models, with participants demanding systems that mirror the fluidity of human decision-making in social interactions.

Opt-out-by-default architectures gathered strong support, with participants challenging industry-standard opt-in models. P03 articulated this preference clearly: “*Data collection should be off until I turn it on for specific things*”, emphasizing the need for explicit user activation of sensing capabilities. This stance reflects broader skepticism toward passive data harvesting practices, as exemplified by P09's critique of deceptive interfaces: “*They give us the illusion of control, but they don't actually let us opt out easily*”. Participants envisioned systems where sensors remain disabled until purposefully activated for defined tasks, contrasting sharply with always-on data collection norms.

Dialogue-based permission systems emerged as a proposed solution for granular control, combining authentication with contextual consent. P06 described a scenario where “*Only listen to these two people... So, if I say, 'show me my bank accounts,' it would recognize my voice and do that, but it wouldn't do it for anyone else*”, positioning vocal biometrics as both a security measure and a consent trigger. This approach addresses participants' concerns about unauthorized access while maintaining functionality, particularly for users with mobility or dexterity challenges. Participants contrasted

this proposed system with existing privacy interfaces, criticizing current settings as “*hidden under layers of menus*” (P05), which create barriers to meaningful control. The voice-based paradigm directly counters the opacity of conventional digital controls, replacing nested configuration screens with intuitive, context-aware verbal commands that align with participants’ demands for “*simple, accessible privacy settings that don’t require an engineer to figure out*” (P01).

These solutions, highlighted by older adult participants, bridge the gap between automated convenience and manual control, enabling personalized privacy management without constant user intervention. By integrating adaptive mechanisms into tangible storage systems, older adults aimed to create responsive yet transparent systems that respect individual autonomy and minimize the complexity of configuration. We also found that each of these participant-generated design solutions touch upon various aspects of physical, psychological, social, and information privacy as listed in Table 4.

5 Phase 2: Speculative Design of Privacy Solutions

Following and extending upon insights from our PD analysis process, we engaged in SD as a *design exploration* to knit together and interconnect older adults’ concerns and design ideas around privacy with the research team’s domain-specific knowledge of social robot design, revealing potential privacy design features for social robots. This process was informed from older adults’ concerns and design ideas from the PD workshops around context-aware privacy controls (Sec. 4.4.4), tangible privacy controls (Sec. 4.4.5), dialogue-based data collection permission systems (Sec. 4.4.6), and personal data security (Sec. 4.4.4-4.4.6). These connections are more specifically described in the following sections.

5.1 Methodology

We developed a set of SD storyboards to facilitate the extraction of social robot privacy design features. The SD storyboards were informed by the PD workshops with older adults (Figure 5), specifically the older adult storyboards created in W1 and the conversations and reflections around the W2 privacy scenarios and related privacy designs. Researchers collectively reviewed the PD recordings and storyboards, systematically identifying older adults’ interaction designs, concerns, and interests. After four iterative rounds of discussion and synthesis among the research team, the team distilled three core design insights and priorities from the PD workshops. These three core designs insights provided the foundation for the SD exploration. In the core design insight sections that follow, we provide further detail into how the design insights connect to older adults’ ideas from the PD workshops (previously described in Sec. 4.4).

Once the three core design insights were collectively defined, the research team ideated various speculative storyboard ideas. One researcher then refined the ideas and generated the speculative storyboard designs. The initial speculative storyboard drafts were collaboratively reviewed in multiple rounds by the research team. Feedback sessions focused on ensuring that the storyboards were representative of older adult concerns and captured the core design

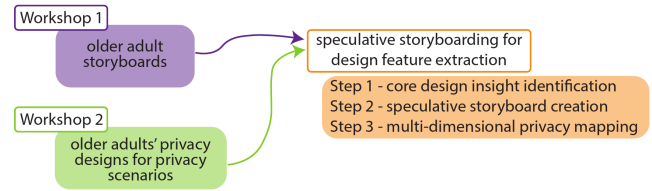


Figure 5: Overview of the speculative storyboarding for design feature extraction. Based on the storyboards from W1 and privacy scenarios from W2, we (1) identified the core design insights for older adult-centered social robot privacy interfaces, (2) created speculative storyboards, and (3) mapped these storyboards to different privacy dimensions.

insights. Visual elements, character behaviors, and privacy mechanisms were iterated and refined to ensure narrative clarity while retaining the exploratory intent of SD. Each speculative storyboard illustrated an interaction involving an older adult and a social robot, incorporating privacy-preserving features older adults discussed in the PD workshops (e.g., local storage, consent prompts, physical controls). Dialogue and visuals in the speculative storyboards were grounded in older adult privacy concerns and proposed solutions.

The finalized storyboards served as tools to support identifying and forming key design features and to understand how they do or do not represent multi-dimensional privacy, mapping to the 4 types of privacy [7, 57]: physical, psychological, social, and informational. This post-PD process of creating the speculative storyboards helped the research team to understand how various design features related to privacy could knit together to support a holistic approach to privacy. Overall, the speculative storyboards created by the research team enabled us to synthesize participants’ privacy values into narrative form and explore how design interventions could address older adults’ core privacy concerns. In the following section, we share each of the core design insights (situational adaptability, physical privacy signaling, and secure multi-model input), demonstrate the speculative storyboard design solution, and describe how the storyboard intersects with the 4 privacy types.

5.2 Core Design Insight #1: Situational Adaptability

This insight emphasizes designing robots to *flexibly* adjust their behavior and data-sharing settings based on the user’s current context or needs. As a concept, situational adaptability empowers users to pre-set or dynamically change privacy and functionality controls based on their situational preferences and comfort. To support situational adaptability, robots should provide context-sensitive privacy configurations. These behaviors should also be easily accessible through voice commands or simple interfaces that could help reduce cognitive load while ensuring the robot’s behavior aligns with real-time needs. This design insight draws from older adults’ design ideas from the PD workshops around preferences for context-aware privacy controls (Sec. 4.4.4), including robots adjusting their data sensing and sharing behaviors depending on the context, activity (such as walking, resting, emergencies, etc.), or location (e.g., public vs. private spaces) and an intuitive interface for data sharing

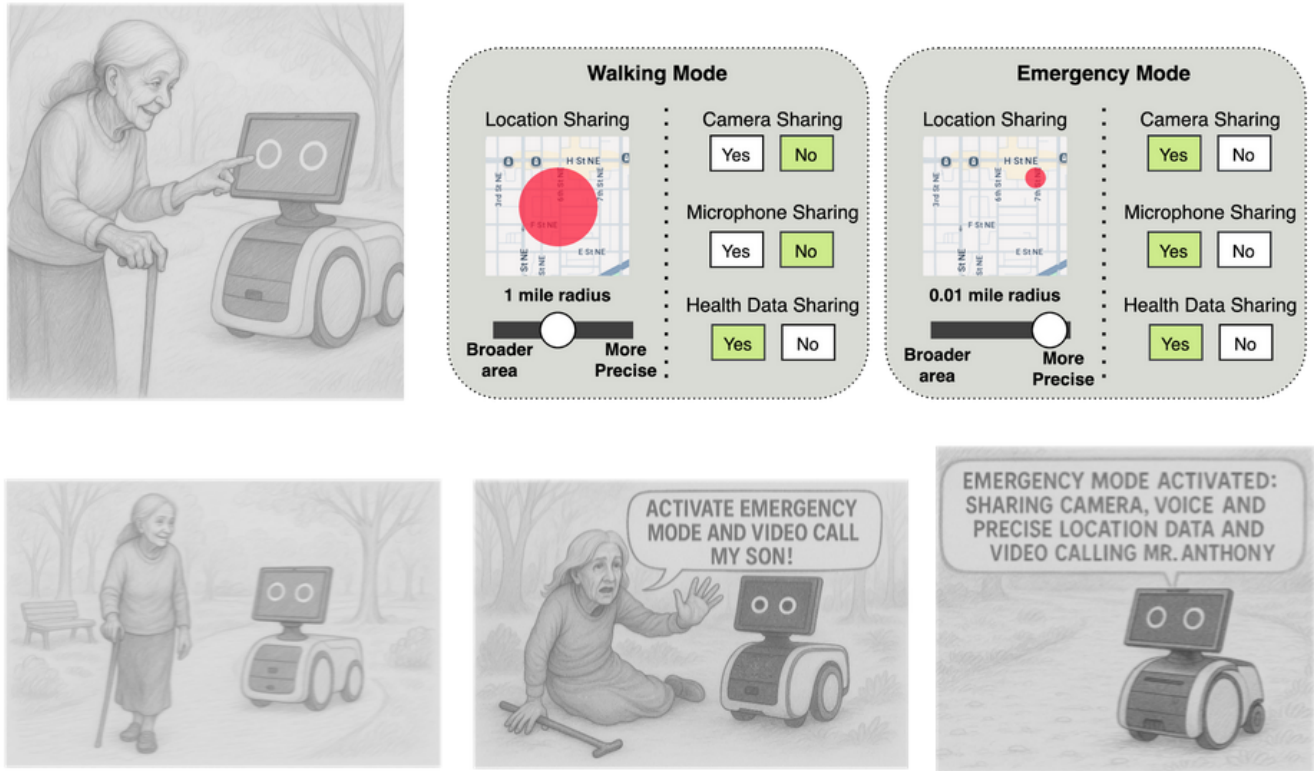


Figure 6: Walking Buddy storyboard illustrating an older adult activating “Walking Mode” on their robot, which adjusts sensing behaviors by disabling the data sharing of camera and microphone, limiting location precision, and selectively sharing health data. The scenario visualizations were created using generative AI.

settings (such as the location slider and mode-based privacy). The scenario of the “Walking Buddy” speculative storyboard (Figure 6) was inspired by P06’s “Walking Buddy” PD storyboard.

5.2.1 Walking Buddy as a Solution for Situational Adaptability. In the “Walking Buddy” speculative storyboard, Dori, an older adult, steps outside with her robot, *Astro*. She opens its interface and selects “Walking Mode”, which automatically applies the following privacy settings based on the context of a casual outdoor stroll:

- Camera & Microphone Sensors: **ON**
- Camera & Microphone Data Sharing: **OFF**
- Health Data Monitoring & Sharing: **ON**
- Location Sharing: **ON (1-mile radius)**

Astro walks quietly beside her, lighting the trail and occasionally pointing out birds, supporting Dori’s safety and comfort without overstepping boundaries. Suddenly, Dori trips and falls. She calls out: “Emergency mode!”. In response, *Astro* immediately reconfigures its privacy settings, sharing its microphone, camera, and location data. It then contacts Dori’s son, Anthony, via an in-built app. He receives her health data and real-time location and quickly comes to assist her.

This storyboard illustrates how participants wanted robots that could flexibly toggle between privacy-preserving and help-enabling

configurations. It emphasizes the need for systems that understand when to minimize data exposure and when to prioritize assistance—without making older adults dig through complex menus. Overall, the Walking Buddy storyboard encapsulates how situational privacy should not be treated as a binary (on/off), but as a spectrum that evolves with user context. This theme recurred in other speculative storyboards as well, where participants sought privacy interfaces that were proactive, meaningful, and easy to manage.

5.2.2 Walking Buddy Mapped to Privacy Dimensions. The Walking Buddy speculative storyboard strongly supports psychological and social privacy by controlling data sharing during normal walks. In “Walking Mode,” the robot’s camera and microphone do not share or store the collected data, giving the user peace of mind that nothing is being recorded. This preserves social privacy as well since conversations with friends or bystanders are not shared or stored. Only minimal data (for example, approximate location within a one-mile radius or basic health stats) is shared when needed, so that information disclosure is kept under the user’s control. As one workshop participant put it, “Will this robot take my picture? ... [Where] would [the] photos be sent?” (P07), highlighting exactly the kind of camera-related concern that the Walking Buddy design avoids by keeping the camera, mic, and exact location data sharing

Table 5: Privacy dimensions supported across speculative storyboards' robot scenarios.

Design Concept	Physical Privacy	Psychological Privacy	Social Privacy	Informational Privacy
<i>Walking Buddy</i>	N/A	User can walk with peace of mind, as the robot is not actively recording/sharing camera / microphone and location data.	Camera, microphone and location data sharing is off during routine walks, preventing social privacy intrusion.	Only essential data (e.g., limited location and health stats) is shared based on context, giving the user control over personal information.
<i>Life Record</i>	N/A	Easy inspection and deletion of recordings gives the user peace of mind that private conversations remain private.	When a visitor interacts, the robot asks the owner's permission before recording, respecting privacy in social settings.	Personal data stays local (no cloud upload), with each person's data kept on a separate card so the user has full control to delete it.
<i>Daily Assistant</i>	The robot uses props (ear-muffs or metal sheets) to cover its camera and microphone. The user can detach sensors from the robot (plug and play) when required.	User can freely have private conversations (e.g., with their doctor) knowing the robot is not listening or recording.	N/A	Sensitive data (e.g., medication schedules) remains protected via voice authentication, preventing unauthorized access.

off in normal walking mode. This is summarized in Table 5 (includes all speculative storyboards).

5.3 Core Design Feature Insight #2: Physical Privacy Signaling

Physical privacy as a core insight supports *transparent* understanding of robot's actions related to privacy. To support this core insight, robots should incorporate tangible, visible, and physical indicators of data collection and storage activities. These mechanisms can enhance user trust by providing immediate, easily understood feedback and allowing for real-time user privacy intervention and control of data collection and storage. This design insight draws from participants' emphasis of the importance of tangible privacy controls (Sec. 4.4.5) and dialogue-based permission systems around data collection (Sec. 4.4.6) in the PD workshops. Older adults envisioned this through methods, such as modular storage media and robots asking permission to collect and/or store data. The scenario of the "Life Record" speculative storyboard was inspired by P05's "Life Record" PD storyboard.

5.3.1 Life Record as a Solution for Physical Privacy Signifiers. In the "Life Record" speculative storyboard (Figure 7), Alex is hosting his friend, John, at home and introduces him to his robot, *Misty*. When John tries to talk to *Misty*, it asks Alex: "A visitor is speaking. Should I activate my camera and microphone?" Alex agrees, but disables cloud storage on the robot as he does not want the conversation to be recorded and stored for later access. Alex and John continue talking. In the conversation, John tells Alex that he wants to share some sensitive news with Alex. Later, after John had left, Alex remembers that John had told him some sensitive news that John wished would remain private. Alex worries that *Misty* might have kept recording during this private conversation. To support John's

privacy, Alex opens *Misty*'s chest panel, revealing labeled SD cards. The labels on the SD cards, read "Alex", "Lila", "Rony", and "Visitor - John". Alex pulls out John's card, plugs it into his laptop, and deletes the data. This allows Alex to maintain full control of what information is stored and how it is stored.

5.3.2 Life Record as Mapped to Privacy Dimensions. The "Life Record" design centers on giving the user tangible control over recordings. Psychological privacy is addressed through easy inspection and deletion of data. Users can open the robot's chest to see and remove stored clips, providing "peace of mind" that private conversations have not been kept without their consent. Social privacy is built in by requiring permission before recording strangers: the robot asks the owner for approval whenever a visitor speaks, so that guests are not recorded automatically. Informational privacy is supported by strictly local storage: all video/audio remains on removable SD cards (one per person) rather than in the cloud. This matches participants' mistrust of cloud systems (one even likened it to "Big Daddy in the sky" (P09)) and lets the user delete any person's data by physically ejecting their card. In sum, "Life Record" keeps recordings tangible and segregated by user, so that the older adult maintains complete control over what is kept or erased.

5.4 Core Design Feature Insight #3: Secure Multi-Modal Input

Through this insight, privacy-sensitive interactions, including accessing health records or managing sensor settings, should be secured via intuitive, multi-modal inputs like voice authentication, facial recognition, physical gestures, or removable hardware (e.g., plug-and-play style; personal tokens such as bracelets that are removable). These multi-modal inputs should be configured based on user privacy concerns and comfort levels, and accommodate diverse

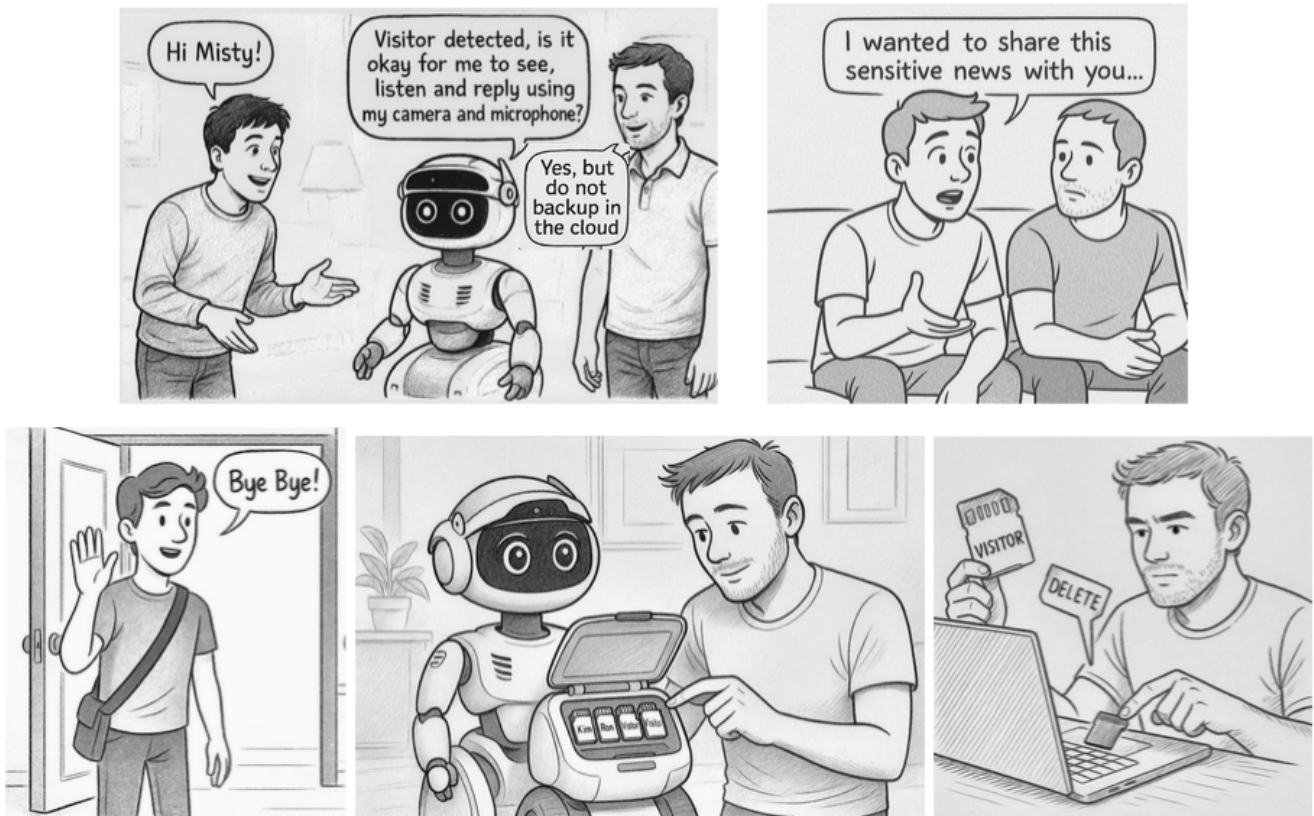


Figure 7: Life Record storyboard showing a robot that asks for the owner’s permission before activating sensors during a visitor interaction. The robot stores data locally on labeled SD cards, allowing the user to manually delete specific recordings belonging to specific individuals. The scenario visualizations were created using generative AI.

interaction styles and accessibility needs. This design insight was informed by older adults’ ideas of how personal data could be kept safe and secure, including: (1) embodied privacy controls or physical mechanisms, such as sensor shielding or detachable components, to support real-time privacy control (Sec. 4.4.5); (2) voice-gated access as voice authentication methods that are non-intrusive and adaptive for managing access to private data (Sec. 4.4.6); and (3) contextual nudges or friendly, timely reminders about upcoming events, paired with situational privacy prompts (Sec. 4.4.4).

5.4.1 Daily Assistant as a Solution for Secure Multi-Modal Input. In the “Daily Assistant” speculative storyboard (Figure 8), Mrs. Selvig, an older adult living independently, interacts with her personal assistant robot, *Misty*. In the morning, she asks, “Misty, can you show me my current medication schedule?” Recognizing that this request involves sensitive health information, Misty initiates voice authentication. When Misty successfully matches the voice to Mrs. Selvig, she displays the medication schedule on her face-mounted screen.

Later in the day, Mrs. Selvig’s housekeeper attempts to access the same information. Misty listens and performs a voice authentication. The process is unsuccessful as the voice does not match Mrs. Selvig’s voice. Misty replies to the housekeeper, saying “Accessing sensitive

data. Voice recognition failed.” and does not provide access to Mrs. Selvig’s medication schedule.

Later in the day, Mrs. Selvig is relaxing at home. At 3:00 PM, Misty gently prompts, “It’s time for your online appointment with Dr. Mark!” While grateful for the reminder, Mrs. Selvig feels uneasy about the robot collecting voice or video data during the call with her doctor. She activates one of Misty’s *Private Modes*, designed to respect her boundaries. Mrs. Selvig’s first option is a privacy mode where Misty will physically cover its camera and microphone, using a mechanical shield and earmuffs. The camera and microphone sensors are turned off and not collecting data while the robot is in this position. The second privacy mode option is Misty’s sensors can be detached from its body (“plug-and-play” style), enabling selective sensing only when reconnected to the robot. Mrs. Selvig decides to choose the first privacy option and Misty enters into this mode while Mrs. Selvig attends her virtual doctor’s appointment.

5.4.2 Daily Assistant Mapped to Privacy Dimensions. The Daily Assistant speculative storyboard uses concrete, physical measures to ensure privacy. Physical privacy is explicitly designed: when the user needs privacy, they can cover the robot’s camera and microphone (for example, by placing earmuffs or a metal shield on it) or even detach the sensors altogether. This also supports psychological

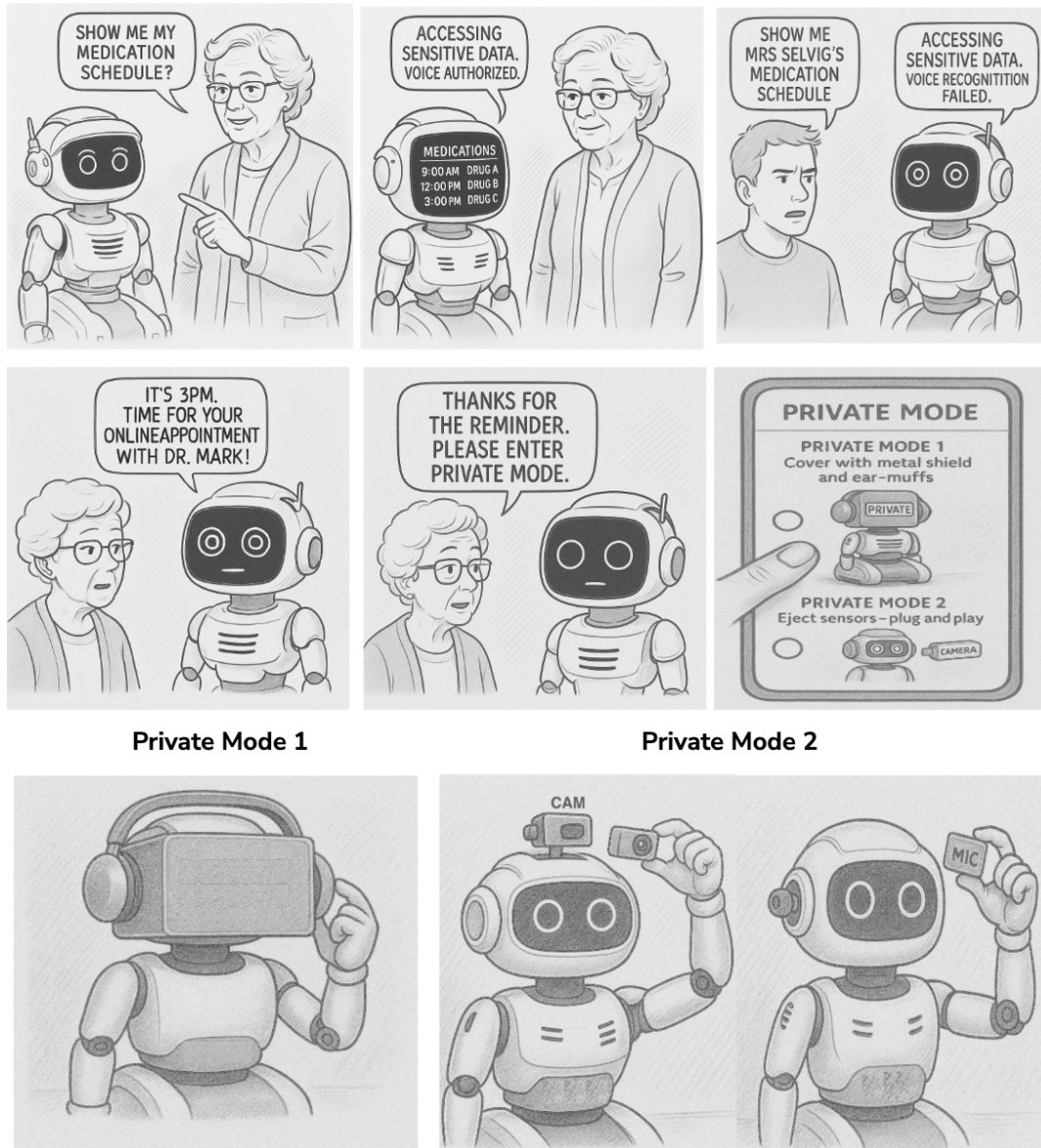


Figure 8: Daily Assistant storyboard in which a robot uses voice authentication to protect sensitive health information from unauthorized access. During a doctor's appointment, the user activates a physical "Private Mode" by either shielding sensors or detaching them entirely from the robot (plug-and-play style). The scenario visualizations were created using generative AI.

privacy: the user can have a confidential conversation (e.g., with a doctor) knowing the robot is not listening or recording. In terms of information privacy, sensitive personal data (like medication schedules) is secured behind access controls – for example, the robot uses voice recognition so that only the authorized user can retrieve private health information.

6 Discussion

Our findings reveal that older adults conceptualize privacy not as a static configuration, but as an evolving, distributed practice

shaped by interpersonal trust, context-awareness, and shared routines echoing prior work on *boundary negotiation*, which frames privacy as a dynamic coordination of access, visibility, and interaction in response to changing social and situational contexts [56]. We elaborate on this focused on privacy over time and privacy across dimensions in the following sections. We also discuss how speculative design can complement PD processes.

6.1 Privacy Over Time

Older adults in our study emphasized that consent and data practices must be continually revisited and renegotiated as privacy is

not a static, one-time agreement but a dynamic, ongoing process that evolves throughout their interaction with the robot. This perspective directly challenges the default design practice in many smart home and robotic systems that treat privacy decisions as a one-time action, e.g., a single consent form or a fixed set of preferences defined at first use [33] and aligns with Palen and Dourish's [56] seminal view of privacy as a dynamic "boundary regulation" process. Such an iterative and revisitable consent process requires easy ways for older adults to modify privacy permissions as their comfort levels with the robot evolve.

While some prior work used AI-based automatic privacy adjustment (e.g., Tang et al.'s Confident [74] system), our participants emphasized the importance of allowing human oversight and consent renewal through naturalist interactions (e.g., asking about privacy settings when contexts shift) and intuitive control of what and when data was collected. This shows older adults' preference for a *collaborative* model of privacy management: the robot can be smart about various contexts, but it must also keep the user in the loop and empower them to set personalized rules. Accessible privacy interfaces that embody this model were illustrated in the "Walking Buddy" speculative storyboard with mode-based privacy adjustments (e.g., "Walking" or "Emergency") alongside access to a settings dashboard to manually configure data sharing and sensor activation. Based on these insights, we suggest that future robots should be treating data collection and privacy consent as an ongoing process for users to easily manage their data privacy in accessible ways.

Older adults' preference for physical privacy controls reflects a broader skepticism toward software-based mechanisms. Older adults desire tangible and intuitive interfaces (e.g., robot earmuffs in the "Daily Assistant's" scenario or detachable sensors) for better transparency and intervenability and noted that such designs would enhance older adults' trust, who may be less comfortable navigating complex digital interfaces [2, 34, 49]. Thus, we suggest designing privacy controls that are both visible and directly manipulable, so that future robot systems can better meet older adults' needs for clarity, agency, and reassurance in everyday interactions.

Participants also framed data deletion as an integral part of privacy over time, rather than a trivial afterthought. Instead of assuming that once data is collected, the matter is closed, older adults in our study indicated continuous monitoring/managing of personal data is essential, which contrasts with common data management practices where data is indefinitely accumulated unless the user performs a manual reset [42, 56]. Older adults' proposed approach for "cleaning data" periodically extends prior research on data protection principles, such as data minimization and user control after collection [40, 42]. Implementing such features would serve as a design provocation – envisioning privacy management not as a static setting but as a temporal workflow.

In summary, older adults view privacy with social robots as a living, breathing arrangement that must be maintained routinely over time. This perspective challenges robot privacy design to move beyond static interfaces and one-time privacy decisions, advocating instead for adaptive systems that support ongoing privacy negotiation [25, 65]. By designing with a sensitivity to temporal privacy dynamics, we can move closer to robots that are not just helpful and intelligent but also trustworthy long-term collaborative systems.

6.2 Privacy Across Dimensions

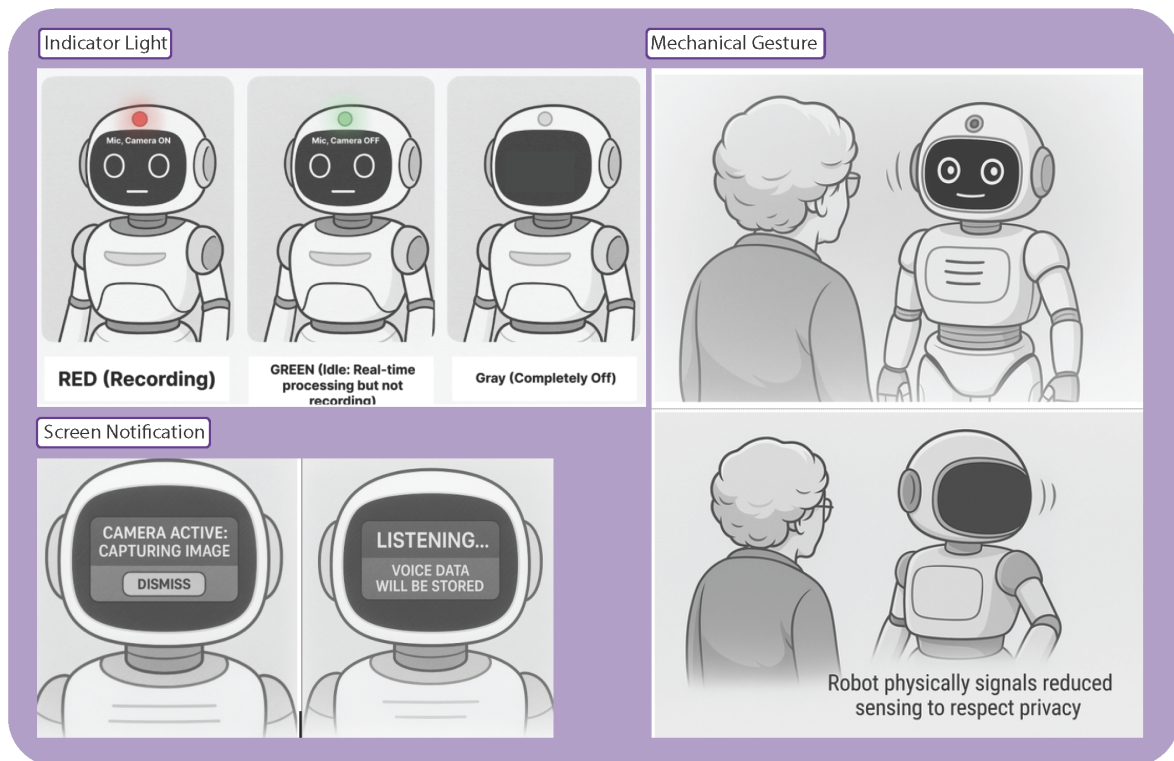
Our findings demonstrate an emerging insight of how older adults conceive of privacy as a holistic, integrated experience encompassing physical, informational, psychological, and social dimensions (Tables 4 and 5). Rather than treating these as separate silos, participants discussed privacy in terms of maintaining an overall sense of personal autonomy and comfort in the presence of a robot. Such narratives illustrate how, in practice, older adults naturally weave physical, informational, psychological, and social considerations into one cohesive understanding of privacy. This is echoed in Palen and Dourish's framework that posits privacy as a dynamic, dialectical process involving multiple interdependent boundaries [56]. This integrative framing of privacy stands in contrast to much prior work, which has often addressed one or two dimensions of privacy as isolated concepts [1, 19, 32, 74]. However, framing privacy solely as a contextual trade-off [19, 32, 74] can overlook the broader interplay of privacy dimensions.

Our work emphasizes how social robots for older adults need to be carefully designed to consider all four dimensions of privacy simultaneously that emerged in our work. Notably, participants often pointed out that a breach in one dimension could undermine privacy as a whole because robots are more than just tools but are often perceived as a *sociable agent*: for instance, if the robot over-shared personal information with others without permission, it would not only violate informational privacy but also make them feel vulnerable and socially exposed, eroding trust (a psychological impact) and making them want more physical distance from the robot. This underscores that the four dimensions reinforce each other in practice. This holistic approach can inform the next generation of HCI privacy research: rather than tackling, say, data security in isolation or social acceptability in isolation, designers can use our insights to jointly address the full suite of privacy needs.

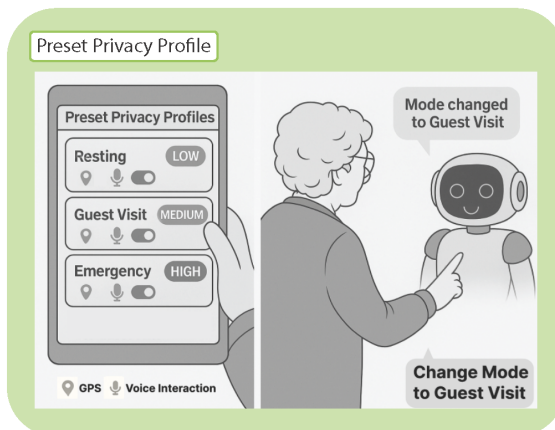
6.3 Speculative Design as a Compliment for Participatory Design

Our SD approach extended the PD data of older adults' storyboards, providing a foundation and context for the research team to understand how older adults imagined social robots to support their privacy preferences through insightful privacy interface features for robots that touch upon various privacy dimensions. Most of these designs were high-level design guidelines and concepts. The SD process explored how these core design insights could support and realize older adults' visions, desires, and values around privacy, centering older adults and incorporating research expertise in social robot design. Similar to Cheon & Khovanskaya's reflections on their use of SD in ethnography [12], our SD process enabled us to consider the PD insights more deeply, revealing the complexities, nuances, and tensions that are considered within privacy design and its physical, psychological, social, and informational influences. This supported us in translating older adults' privacy design ideas to detailed design features that embodied older adults' values. Our speculative storyboards incorporate older adults' values and position the older adult having autonomy of the technology, critiquing ageist perspectives around older adults' relationship with technology [23] as a commitment in critical design [4, 79].

REAL-TIME PRIVACY FEEDBACK



PRIVACY PROFILE



DATA OWNERSHIP TOOLS

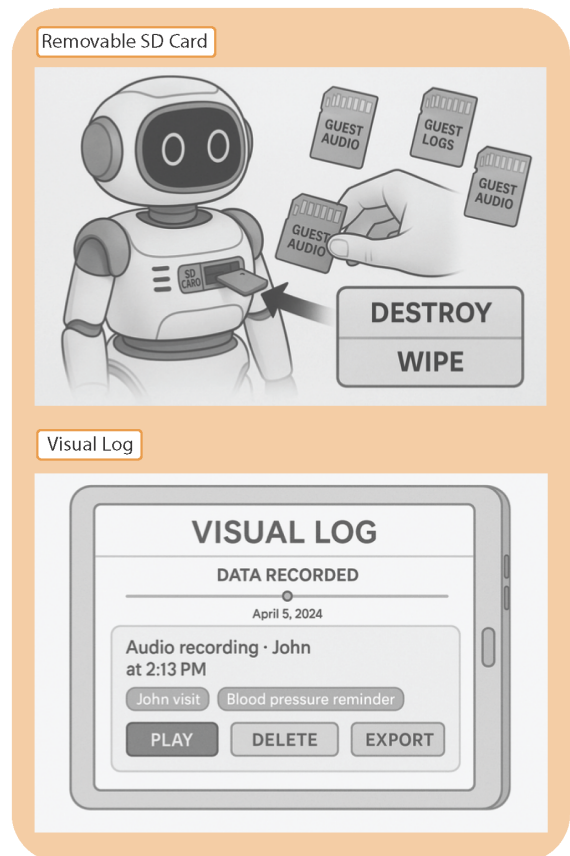


Figure 9: Three design features synthesized from PD workshops with older adults and SD extensions of workshop storyboards in the areas of real-time privacy feedback, privacy profile, and data ownership tools. The scenario visualizations were created using generative AI.

As we move closer to adaptive privacy system design, we can continue to explore, critique, and modify these social robot design features elicited through SD to contend with how these designs may interact with existing technologies and infrastructures, such as privacy regulation (e.g., GDPR). This is emblematic of the complexity of robot design dedicated to designing socially responsible systems [68].

7 Design Recommendation: Context-Aware, Customizable Privacy Modes

Our post-PD SD process demonstrates another way that SD approaches can be used to engage and reflect by the research team in HCI/HRI design processes, expanding from previous work [12, 18, 75]. This process was critical in reflecting older adults' proposed design features and perspectives/concerns into detailed interactions and inspiring detailed features that would support older adults envisioning robot interactions.

Building on themes identified across our SD storyboards, we recommend that future social robot systems should reflect how privacy preferences shift over time and across settings and give users tools to manage those shifts fluidly. Privacy should not be handled as a single, binary choice, but instead managed as an ongoing part of daily interaction—responsive to situational, emotional, and social cues. To realize these systems, we propose three interrelated design features drawing from the SD storyboards that could support dynamic, situational privacy in robots: preset privacy profiles, real-time feedback, and data ownership tools (Figure 9). Collectively, these three design features offer a multi-layered approach to robot-mediated privacy, as the design features can be combined to promote a holistic approach to privacy design.

7.1 Preset Privacy Profiles

We recommend that robots should support a range of pre-configured privacy modes, such as “Resting,” “Guest Visit,” or “Emergency.” Each profile could correspond to distinct combinations of sensing and data-sharing behaviors (e.g., camera off, health data on, or location sharing limited to emergency contacts). These profiles should be accessible through multiple modalities (i.e. voice command, touchscreen menu, or physical interface) allowing users with different abilities to adjust their privacy with ease. This feature centers on a simplified, context-based privacy management system, allowing users to choose from pre-configured modes. These modes are presented via a user-friendly interface—typically on a tablet or touchscreen display—and are also accessible through voice commands or physical toggle switches to accommodate different user preferences and capabilities.

Each privacy mode should clearly indicate which sensors are active (e.g., camera, microphone, location) using visual toggle icons. Additionally, each profile should be labeled with an associated privacy level (e.g., Low, Medium, High), which helps users quickly assess their data exposure. To reinforce transparency, real-time system feedback should also be provided (explored in more detail below). By integrating privacy controls into contextualized presets, this approach minimizes cognitive and operational burden while supporting predictable and explainable privacy behavior in the robot.

7.2 Real-Time Feedback

We also recommend that the robots provide real-time notification to users on which sensors are being used and what types of data are being collected for full transparency at any given moment. Visual cues, such as indicator lights, screen notifications, or mechanical gestures (e.g., a robot turning away), could provide immediate visibility into the robot's current privacy state, reduce uncertainty, and help maintain older adult users' psychological comfort, assuring that privacy boundaries are being respected.

For instance, privacy indicator lights, such as a light band or dot on the robot's body (e.g., head or chest), could change color based on sensing activity (e.g., red for recording, green for idle, and off for inactive/private modes). We draw inspiration from conventional privacy cues like webcam lights. Another design for real-time feedback could be screen-based notifications; the robot's display presents clear textual messages such as “Camera Active: Capturing Image” or “Listening...Voice data will be stored.” Where possible, users could dismiss or acknowledge these notifications, creating a sense of interaction and affirmation. Lastly, embodied robot gestures could be used to indicate any privacy transitions. For instance, a robot could turn its head or body away, or close camera shutters, signaling that it is disengaging from active sensing. Real-time feedback mechanisms could promote transparency for users who may be unfamiliar with digital systems to visibly validate the robot's behavior.

7.3 Removable Memory Cards for Various Personal Data

Participants consistently emphasized a desire for tangible control over their data. To meet this need, we recommend that robots store data locally and provide visible and easily accessible mechanisms for users to review and manage any collected data. We speculated that a robot's body would include visible and labeled ports that contain distinct memory cards for various personal/interaction data (e.g., “Health Logs,” “Guest Audio”). Users can physically remove or destroy these cards, providing a tangible mechanism for data management.

Another way users could manage data is by managing event-based visual data logs. The robot may use a screen interface that displays a chronological timeline of recorded events that the user could review. Each data entry point could include options, such as “Play,” “Delete,” or “Export,” with the ability to tag data such as “John's Visit” or “Blood Pressure Reminder.” This interface could help users understand when, why, and what data was recorded and offers control over future data use. Providing both physical and digital means for reviewing and removing data reinforces ownership and consent in ambient data collection and supports understanding and managing data tailored for different groups of users.

These design features reflect a broader understanding of privacy as situational, experienced, and adaptive; not merely agreed to at a single point. Designing for this more dynamic conception of privacy has the potential to improve long-term user comfort and enable greater adoption of social robots in aging-in-place contexts.

Our work is not positioned to evaluate design feature feasibility as it is outside the scope of our SD process. However, the design features are one embodiment of how dynamic and situational privacy

can be designed in social robots that we can use as design directions to guide and inspire future robot design in HCI and HRI. Various design features, such as screen notification of data collection and preset privacy profiles, may be implementable with current technology constraints. While other design features, such as removable SD cards to facilitate data deletion, may require larger technological changes. This also applies to how these design features will interact with existing privacy legislation. Overall, by extending the PD workshop through SD, we were able to build upon older adults' design ideas to develop specific and detailed design features that promoted open-exploration of how social robot design can support older adults' privacy through a multidimensional lens.

8 Limitations and Future Work

As qualitative work provides a foundation for understanding and providing descriptions around human behavior, our work's findings are not meant to be generalizable, but instead provide depth and description around how older adults design around privacy and how SD can build upon PD. One limitation of our work is the homogeneous sample of fourteen older adults living in assisted living facilities from a single region (the Midwest United States) that participated in the PD workshops. Future research should explore privacy design for social robots with more diverse cohorts from varied regions, countries, and ethnic, socioeconomic, and technological backgrounds. This will support greater understanding of how culture and other demographic aspects may influence older adult privacy perceptions. Additionally, these workshops could be expanded to include caregivers, family members, and co-residents. Next, future work includes translating the generated privacy design ideas into functional robot prototypes and/or simulations, enabling long-term field studies and providing insight into how older adults interact with and adjust privacy mechanisms over time.

9 Conclusion

Our study provides new insights into how older adults perceive privacy of social robots. Rather than viewing privacy as a static, one-dimensional setting, participants framed it as a dynamic, evolving phenomenon shaped by experience and context. They articulated privacy across four interrelated dimensions—physical, psychological, social, and informational—highlighting the need for robots that respect these boundaries in tandem. These insights call for future work that recognizes privacy as multifaceted and socially embedded, evolving with the user's relationship to the technology. Our work also makes a methodological contribution by using SD as a post-PD research tool. The generated speculative storyboards and subsequent feature extraction interpreted, realized, and built around older adults' privacy concerns for social robot, merging older adults' and the research team's expertise. This narrative-driven approach modeled how to incorporate privacy concerns, robot interactions, and privacy design based on older adults' PD designs and considerations. Our results carry concrete implications for future robot design and privacy research. We propose three design features that support adaptive privacy mechanisms in social robot design that allow users to configure privacy settings based on situational needs and personal comfort.

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